Program 1: Sum + Decision (input + conversion + if/elif/else)

Goal: The program reads two numbers, calculates the sum, prints it, and prints an extra message depending on the sum.

Instructions (steps)

1. Create a file named `program1_sum_decision.py`.
2. Ask the user for the first number using `input()`, and convert it to a number using `float()`.
3. Ask the user for the second number the same way.
4. Add the two numbers and store the result in a variable named `s`.
5. Print the result using an f-string: `Sum: ...`
6. Use `if / elif / else`:
   - If `s > 20`, print `Large sum`
   - If `s == 20`, print `Exactly twenty`
   - Otherwise, print `Small sum`
7. Test your program with these inputs: `(10, 15)`, `(7, 13)`, `(5, 2)`.

Program 2: Safe List Total (while + try/except + list + append + for)

Goal: The program safely collects 5 numbers from the user (no crashing), stores them in a list, and prints the total.

Instructions (steps)

1. Create a file named `program2_safe_list_total.py`.
2. Create an empty list: `numbers = []`.
3. Use a `for` loop with `range(5)` to collect 5 values.
4. Inside the loop, use `while True` + `try/except` to safely read one number:
   - Try to read a number with `float(input(...))`
   - If it works: add it to the list with `numbers.append(x)` and exit the loop with `break`
   - If it fails (`ValueError`): print an error message and ask again
5. After collecting 5 numbers, calculate the total:
   - Start with `total = 0`
   - Loop through the list: `for x in numbers: total += x`
6. Print the result using an f-string: `Total: ...`
7. Test your program by typing: `2`, `abc`, `3`, `4`, `5`, `6` (it should not crash).

Program 3: Find the Biggest Number (list + for + if)

Goal: The program asks the user for **5 numbers**, stores them in a list, and prints the **largest** number.

Instructions (steps)

1. Create a file named `program3_biggest_number.py`.
2. Create an empty list: `numbers = []`.
3. Use a `for` loop with `range(5)` to read 5 numbers:
   - Use `float(input(...))`
   - Add each number to the list with `append()`
4. Set `max_value` to the first element in the list: `max_value = numbers[0]`.
5. Loop through the list:
   - If a number is greater than `max_value`, update `max_value`.
6. Print the result: `Biggest: <value>`
7. Test with inputs like: `3, 10, -2, 7, 7`.

Program 4: Safe Sum Until Stop (while + try/except + if + running total)

Goal: The program keeps asking the user for numbers and adds them to a total **until the user types `stop`**.

Instructions (steps)

1. Create a file named `program4_sum_until_stop.py`.
2. Create a variable `total = 0`.
3. Start an infinite loop: `while True:`.
4. Ask the user for input: `text = input("Type a number or 'stop': ")`.
5. If the user types `stop` (case-insensitive), break the loop.
   - Tip: use `text.lower() == "stop"`
6. Otherwise, try to convert the input to a number:
   - If conversion works, add it to `total`
   - If conversion fails, print an error message and continue
7. After the loop, print: `Total: <value>`
8. Test: `5, 2, abc, 3, stop`

Program 5: Average of Numbers (list + for + sum + division)

Goal: The program asks the user for **5 numbers**, stores them in a list, calculates the **total** and the **average**, and prints both.

Instructions (steps)

1. Create a file named `program5_average.py`.
2. Create an empty list: `numbers = []`.
3. Use a `for` loop with `range(5)` to read 5 numbers:
   - Convert each input to a number with `float()`
   - Add it to the list with `append()`
4. Calculate the total using a loop:
   - Start with `total = 0`
   - For each `x` in `numbers`, do `total += x`
5. Calculate the average:
   - `avg = total / len(numbers)`
6. Print:
   - `Total: ...`
   - `Average: ...`
7. Test with simple inputs like: `10, 20, 30, 40, 50`.

Program 6: Count Big Numbers (while + list + if + counter)

Goal: The program asks the user for numbers until the user types `stop`. It stores valid numbers in a list and counts how many of them are **greater than 20**.

Instructions (steps)

1. Create a file named `program6_count_big.py`.
2. Create:
   - an empty list `numbers = []`
   - a counter variable `count_big = 0`
3. Start a loop: `while True:`
4. Ask the user: `text = input("Type a number or 'stop': ")`
5. If the user types `stop` (case-insensitive), break the loop.
6. Otherwise:
   - Try to convert `text` to a number with `float(text)`
   - If conversion fails, print an error message and continue
7. If the number is valid:
   - add it to the list with `append()`
   - if it is greater than 20, increase the counter: `count_big += 1`
8. After the loop, print:
   - `Numbers entered: <how many>`
   - `Greater than 20: <count_big>`
9. Test: `5, 25, abc, 21, 20, stop`